AI Stocks, Utility Betas and Allowed Returns: Evidence from Six Regulatory Regimes

Arian Nasrollahi Shiraz

Working draft, 8 November 2026. All numbers come from the scripts in code/ and the tables in output/tables/; the tests were written down in prespecification.md before any utility returns were analysed.

Abstract

Regulators set utilities' allowed return on equity partly from market betas, so a change in how utility shares move with AI stocks could reach electricity bills. I test this channel in six countries with tests fixed in advance. US utilities that disclose more data-centre activity did not become detectably more correlated with semiconductor stocks after November 2022. Rebuilding each regulator's beta with AI-era co-movement held at its pre-2023 level, the AI era adds 6 basis points to the UK's and Spain's 2026 to 2031 allowed returns and 22 to 35 to a prospective German estimate, almost all through the market index. These numbers roughly halve when the counterfactual holds a different moment of the index fixed, and placebo break dates in 2015 to 2021 typically produce contributions as large. In 85 US rate cases, pass-through is not distinguishable from zero.

1. Introduction

The allowed return on equity is one of the largest single parameters in a network bill. Dunkle Werner and Jarvis (2026) estimate that US allowed returns above several capital-cost benchmarks cost consumers an average of $7 billion a year, and Rode and Fischbeck (2019) document a large and growing premium of allowed returns over the riskless rate across nearly four decades of US electric rate cases. In most of the regimes studied here, part of the allowed return comes from the capital asset pricing model (CAPM): the regulator measures how much the shares of comparable listed utilities move with the market (their beta) over a recent window and multiplies it by a market risk premium. Anything that changes measured betas therefore changes allowed returns at the next review, and customers pay the difference whether or not a data centre connects near them.

The AI investment boom could move these betas in two ways. Utilities that expect large data-centre loads might start to trade with AI stocks, which would raise their covariance with the market. And because betas are measured against a local index, a change in how much that index moves with AI stocks changes the beta of any utility whose own AI co-movement differs from the market's. This paper asks whether either channel has reached measured betas since 30 November 2022, and whether it has reached allowed returns.

The answer is that regulatory betas absorbed a small change, that it does not run through utilities' own exposure, and that it is not clearly specific to the AI era. Across the 33 US regulated electric utilities, the change in each utility's loading on an AI factor between 2017 to 2022 and 2023 to 2026 shows no detectable relation to its data-centre exposure: the slope is −0.003 per data-centre mention per 10,000 words of the annual report (standard error 0.028), and 70% of 1,000 placebo break dates give a slope at least as large. The data cannot rule out a modest relation: the 95% interval, about −0.06 to +0.05, admits a gap between the most and least exposed utilities of up to about one cross-utility standard deviation of the

change in either direction. Regulated utilities load negatively on the AI factor once the market is held fixed, about −0.28 before and −0.27 after. Merchant generators moved from −0.32 to +0.46, consistent with investors pricing AI demand into generators that sell power at market prices.

Regulator-style betas fell between windows ending in November 2022 and windows ending in October 2026 in five of the six countries: in the US by 0.46 on a 5-year weekly window, and by 0.17 to 0.31 in Germany, Spain, Italy and Canada. The UK's 10-year beta barely moved. An exact decomposition shows that utilities co-moved less with both the AI-linked and the other parts of the index, while the third term mainly records lower market variance once the 2020 crash leaves the window.

To measure what AI-era co-movement contributed to actual decisions, I rebuild each regulator's beta with its own comparators, window, return frequency and de-levering, then recompute it with AI-era returns put back on their pre-AI footing. The AI era adds 0.006 to the asset beta behind the UK's RIIO-3 determination, or 6 basis points on the allowed return on equity. It adds 6 basis points in Spain and 22 to 35 basis points to a German estimate made in October 2026 for the next electricity period. Putting each comparator's own AI loading back at its pre-AI value changes these betas by at most 0.002. Almost the whole contribution comes from the market index. In the AI era the slope of each index on semiconductor returns fell (for the S&P 500, from 0.47 to 0.34). For the S&P 500 this happened because semiconductor stocks became more volatile on their own: their variance rose by 54% and their covariance with the index by 11%, while the share of index variance they explain stayed at 0.62. In the European indices the covariance itself fell. Holding the slope at its pre-AI value, as the main counterfactual does, gives an index that moves more with semiconductor stocks than the actual one; utilities move against the AI factor once the market is held fixed, so their beta against that counterfactual index is lower. The contributions are larger where more of the window falls in the AI era, from 1 of 6 years in Spain to all of a German 1-year or 3-year window, though three determinations cannot establish this.

Two checks written down after the first results, and before they ran, qualify these numbers. Holding the semiconductor share of index variance at its pre-AI value, instead of the slope, roughly halves every contribution: 3.1 basis points in the UK, 3.4 in Spain, 11.9 to 16.3 in Germany, and about zero in the US, where that share did not change. And the AI-era contributions are typical of what other periods produce. Month-block bootstrap intervals exclude zero under the main counterfactual (UK 1.5 to 11.8 basis points), so the contribution is well measured given the break date. But in 87% of 1,000 placebo break dates between 2015 and 2021 the UK contribution is at least as large, and in 26% to 58% for the German windows. Spain comes closest (5% of 585 usable breaks). The index's slope on semiconductor returns falls after most break dates, so a positive contribution of this size is not a signature of the AI era.

In 85 US rate decisions, I rebuild at each order date a beta built like the Value Line betas common in rate-case testimony (5 years of weekly returns, two-thirds weight on the raw beta), measured against the S&P 500. The pre-specified regression of the change in authorised return on equity on the change in the AI-era contribution gives a coefficient of 2.65 return points per unit of beta (standard error 1.74, wild-bootstrap p = 0.29). The estimate is not significant in any of the pre-specified checks, and the regressor is mostly a function of the decision date: the AI-era share of the beta window explains 80% of its variation across AI-era decisions. With fixed effects for the year of the later decision, the coefficient is −0.12 (standard error 2.32). The money involved is small: about £0.27 million a year per £1 billion of UK regulatory asset value, and at most a median of *12.60 per US customer a year even at full CAPM weight. Both US figures use a rate-base proxy that overstates stated rate base for most units, and the estimated coefficient, which is not significant, implies* 5.60.

This paper relates to three literatures. Work on rate-of-return regulation (Rode and Fischbeck 2019; Dunkle Werner and Jarvis 2026) studies how allowed returns track the cost of capital, but treats the inputs regulators use as given. Van Nieuwerburgh (2026) estimates rolling betas for data-centre real estate during the AI buildout, but does not study utilities or rate cases. A companion event study (Nasrollahi Shiraz 2026) finds that on 27 January 2025, when DeepSeek's R1 model raised doubts about AI electricity demand, merchant generators fell about 19% and ended 20.5 points below regulated electric utilities. Because the main effect found here runs through the market index, the paper also relates to the old point that a beta is defined only relative to a chosen market proxy (Roll 1977); in regulation, the index and window a regulator chooses set how far changes in the index's composition reach allowed returns. Searches of the web and Google Scholar on 7 November 2026, repeated on 8 November 2026, found no study linking AI exposure to utilities' measured betas or allowed returns. Relative to this work, the paper makes three contributions. It measures the AI-era component of the betas that actual determinations used, rather than of a generic beta. It separates the utilities' own AI exposure from the composition of the index, which turns out to carry almost all of the effect. And it traces the effect across borders through regulators' foreign comparators: 2.5 of the UK's 6 basis points come from Spanish and Italian networks.

Section 2 describes how each regulator uses beta. Section 3 describes the data, Section 4 the tests and Section 5 the results. Section 6 reports the pre-specified checks, Section 7 what the results do not show, and Section 8 concludes.

2. How regulators use beta

A regulator using the CAPM sets the cost of equity as $r_e = r_f + \beta_e \times MRP$, where r_f is the risk-free rate, MRP the market risk premium and β_e an equity beta. Because comparators carry different debt loads, regulators first convert each comparator's measured equity beta into an asset beta (the beta of the business with no debt), average the asset betas, and convert the average back to an equity beta at a notional gearing. This step is called de-levering and re-levering. The return on equity therefore responds to a change in comparator betas in proportion to the premium and inversely to the notional equity share.

Table 1 summarises the determinations in scope. All values come from the regulators' own documents, with page references in regulator_methods.md, except the US and Canadian rows, which rest on company filings and secondary sources (for Value Line's method, a utility's description in Kentucky PSC Case 2007-00008, response to data request AG 1-98).

Table 1. Beta in six regulatory regimes

	Decision	Comparators	Window and frequency	Adjustment and de-levering	Published value
UK	Ofgem RIIO-3 Final Determinations, December 2025 (2026 to 2031)	Nine: three UK water companies, National Grid, and five Spanish and Italian networks	10 years, daily (2 and 5 years shown)	No adjustment; Harris and Pringle (1985) de-levering, debt beta 0.075, notional gearing 55% for electricity	Asset beta 0.375; MRP 4.6% (CPIH-real)
Spain	CNMC Circular 9/2025, December 2025 (2026 to 2031)	34 European utilities	6 years to December 2023, weekly	Modigliani-Miller unlevering, Hamada (1972) re-levering, gearing 46%	Equity beta 0.75; MRP 4.98%

Germany	BNetzA; next electricity period from 2029	Foreign network operators; twelve in the July 2026 expert report	1, 3 and 5 years, daily, most weight on 1 year	Vasicek (1973) adjustment (not applied in the replica; Section 7)	Prospective only
Italy	ARERA Delibera 513/2024, November 2024 (2025 to 2027)	Comparators in similar sectors, liquidity filter	5 years (2019 to 2023), daily	Blume (1971), possibly partial; Hamada	Asset betas 0.370 and 0.400, unchanged from 2022 to 2024
US	State commission rate cases	Proxy groups of US utilities	Value Line: 5 years of weekly changes against the NYSE Composite	Adjusted toward 1, about two-thirds weight on the raw beta	Case by case; CAPM is one of several models
Canada	OEB and AUC formulas	None	None	None	Return tied to bond yields; no beta

Two features of Table 1 shape the tests. First, regimes differ in how much AI-era data their betas can absorb: Spain's window contains one AI-era year out of six, the UK's about 2.3 out of 10, and Canada's formula contains no beta at all. Second, comparator sets cross borders, so a change in Spanish or Italian network betas can move the UK's allowed return.

3. Data

Prices. Daily adjusted closing prices from Yahoo Finance for 2 January 2012 to 6 November 2026: 33 US regulated electric utilities, National Grid and SSE (UK), E.ON (Germany), Redeia (Spain), Terna (Italy), four Canadian utilities, US water and gas utilities as placebos, two merchant generators, each regulator's comparators, the local market indices, the VanEck semiconductor ETF (SMH), Nvidia and exchange rates. Returns are log returns in local currency.

Interest rates. Daily 10-year government yields from FRED, the Bank of England, the Bundesbank, the Banco de España and the Bank of Canada; for Italy, the ECB's euro-area series (no official daily Italian series was found). The US Baa corporate spread over the 10-year Treasury (FRED BAA10Y) is the credit-spread control in the rate-case panel.

Data-centre exposure. For US utilities, mentions of data centres per 10,000 words of the main text of the FY2023 10-K, counted under a rule frozen before any returns were analysed (exposure_rule.md). The most exposed utility has 0.98 mentions per 10,000 words. Outside the US, where no country-level data-centre share of demand is published for these five countries, exposure is an indicator for utilities serving a major data-centre hub.

Rate cases. For each of the 33 US utilities, the rate unit is the operating company and state with the largest 2023 retail deliveries in EIA Form 861, under a rule frozen in advance (ratecase_rule.md). Authorised returns on equity, order dates and equity ratios were coded from the utilities' SEC filings: 109 decisions between 2014 and 2026, of which 85 enter the panel. The others are excluded under written conventions (rehearings, formula-rate and plan updates, rate freezes, settlements that state no return, a return set by statute) or still pending (ratecase_coding_notes.md). Every coded decision was checked by hand against the source sentence; 9 of the 104 checked decisions were corrected, three of them in the return itself and the rest in dates or equity ratios. The panel covers 32 utilities in 24 states; 53 pairs of consecutive decisions in 18 states identify the panel regression. Mean authorised returns are 9.78% for decisions before 2023 and 9.70% from 2023 on.

Rate base and customers. Stated rate base where a filing gives it (six units); otherwise net electric plant from FERC Form 1, through the PUDL database, which overstates stated rate base by about 1.2 to 2.7 times in the five units where both exist and is used as an upper bound. Customers are 2023 counts from EIA Form 861.

Regulator inputs. Comparator betas from Ofgem's Draft Determinations (Table 16), the CNMC's explanatory memorandum and the July 2026 expert report for BNetzA (Frontier Economics, Table 3); parameters from the final decisions. Raw downloads are archived with SHA-256 checksums, and every dataset is documented in a script-built workbook (data/project_data_workbook.xlsx).

4. Tests

All tests below were frozen on 7 November 2026, before utility returns were analysed. Three amendments, logged with dates and reasons in the pre-specification, replaced unavailable inputs (the Italian yield series, a country-level exposure share and utility bond spreads); later implementation choices are dated in analysis_notes.md.

AI co-movement (H1). For each utility, I estimate on daily returns

$$r_{u,t} = a + b_m m_t + b_{ai}\, ai_t^{\perp} + b_r \Delta y_t + e_t$$

where m is the local index return, ai⊥ is the SMH return in local currency orthogonalised to the index within the window, and Δy the change in the 10-year yield. The pre-AI period runs from January 2017 to November 2022 without February to June 2020; the AI era from January 2023 to October 2026. The test regresses the change in b_ai across the 33 US utilities on data-centre exposure. A positive slope would mean exposed utilities became more tied to AI stocks.

Beta decomposition (H2). With C = cov(u, m) split into C_ai, the covariance with the part of the index explained by SMH, and C_o, the rest, and V = var(m), the change in beta between windows a and b is exactly

$$\Delta\beta = \frac{\Delta C_{ai}}{V_b} + \frac{\Delta C_o}{V_b} + C_a\left(\frac{1}{V_b} - \frac{1}{V_a}\right)$$

The first term is the exposure channel, the second the non-AI channel, and the third reflects the index's variance. Each country's beta is computed in its regulator's window and frequency, ending 29 November 2022 and 31 October 2026.

Counterfactual replicas (H3). For each determination, I reproduce the regulator's beta and then recompute it with two counterfactual changes on AI-era dates:

- exposure: each comparator's return becomes $r - (b_{ai}^{AI} - b_{ai}^{pre})\ ai^{\perp}$, which puts its AI loading back at its pre-AI value;
- composition: the index return becomes $m - (g^{AI} - g^{pre})\ smh$, where g is the slope of the index on SMH, which puts that slope back at its pre-AI value.

The AI-era contribution is the replica beta minus the counterfactual beta with both changes, re-levered at the regulator's notional gearing and multiplied by its premium, in basis points of allowed return on equity. For the UK and Germany, each comparator's gearing is backed out from the regulator's own published

asset beta and my equity beta, and held fixed in the counterfactual; the check on the replica is whether the implied gearing is plausible (between 0.2 and 0.75). For Spain, the check compares my levered betas with the CNMC's.

Further checks. Three checks were written down after the first results and before they ran (analysis_notes.md). First, because the slope g also falls when SMH becomes more volatile on its own, a second composition counterfactual holds instead the share of index variance explained by SMH (the R^2 of the index on SMH) at its pre-AI value, keeping the AI-era residual variance of the index. Second, each contribution gets a 95% interval from 1,000 bootstrap draws that resample calendar months within the two loading periods (the same months for every comparator), and a placebo rank from the 1,000 break dates used for H1, with the regulator's window keeping its length and its position relative to the break. Third, for the US panel, I measure how much of the regressor is explained by the share of each decision's beta window that falls after November 2022.

US panel. At each decision date, I compute each utility's adjusted beta (0.67 × raw beta + 0.33) on 5 years of weekly returns against the S&P 500 (Value Line uses the NYSE Composite, which is not in the data), and the same beta under the counterfactual; their difference c is the AI-era contribution. For consecutive decisions k − 1 and k of unit u in state s,

$$\Delta ROE_{u,k} = \alpha_s + \theta\,\Delta c_{u,k} + \gamma_1 \Delta y^{10}_{u,k} + \gamma_2 \Delta baa_{u,k} + \varepsilon_{u,k}$$

with the 10-year Treasury yield and the Baa spread averaged over the 30 days before each order. Standard errors are clustered by state, and p-values come from a wild cluster bootstrap with Webb (2023) weights. The pre-specification named clustering by utility; the switch to states was recorded before the panel ran, and both are reported. H3 predicts θ between zero and the market risk premium (about 6 points per unit of beta), because commissions weight the CAPM only partly; New York, for example, gives it one third.

Money (H4). Basis points times the equity share of the regulated asset base, per year and per customer.

5. Results

5.1 No detectable link between data-centre exposure and AI co-movement

Figure 1 plots the change in each US utility's AI loading against its exposure. The slope is −0.003 (standard error 0.028); the fitted gap between the most exposed utility and one with no mentions is about −0.003, against a cross-utility standard deviation of the change of about 0.06. The 95% confidence interval runs from about −0.06 to +0.05; since the most exposed utility has 0.98 mentions, the data admit a gap of up to about one standard deviation in either direction, so the result shows no evidence of a relation without ruling one out. The pre-specified second outcome, the change in the AI factor's partial R^2, gives the same answer (slope 0.012, standard error 0.013). In 70% of 1,000 placebo break dates between 2015 and 2021 the slope is at least as large in the predicted (positive) direction (Figure 2); neighbouring dates share most of their windows, so this is a rank among a few distinct periods rather than 1,000 independent draws. US water and gas utilities, none of which mention data centres, moved like unexposed electric utilities (placebo indicator −0.016, standard error 0.036). Pooling all six countries with exposure standardised within each gives a slope of −0.0015 (wild-bootstrap p = 0.28). With country fixed effects,

Germany, Spain and Italy, which have one utility each, contribute nothing to this slope, so it rests on the US, the UK and Canada, and the bootstrap over six country clusters is fragile.

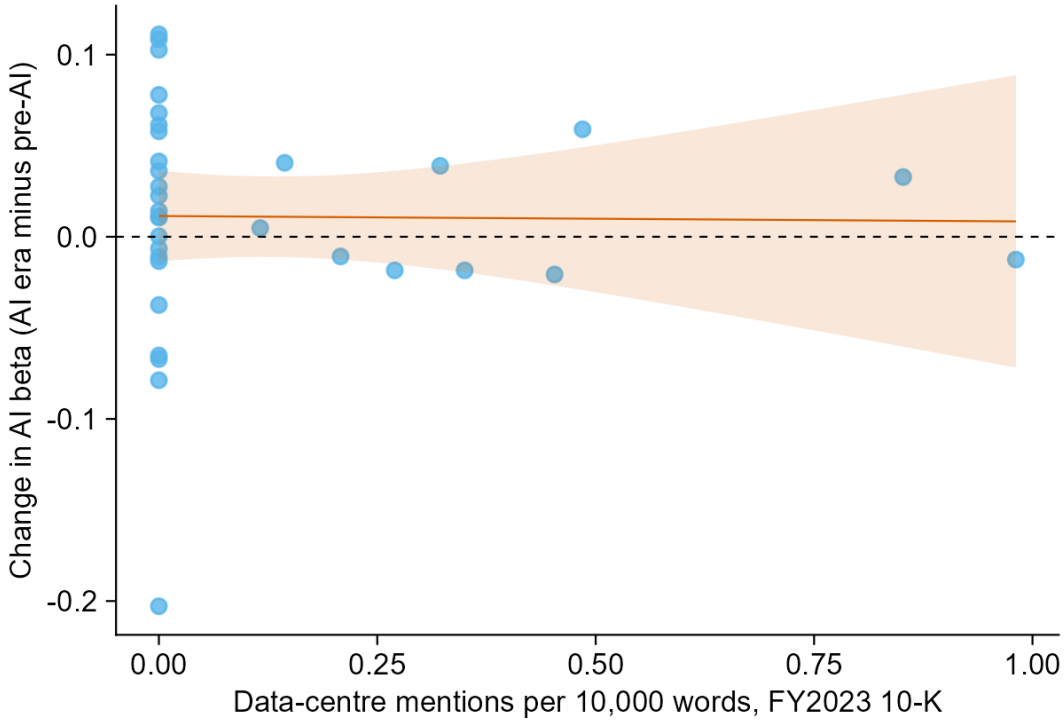


Figure 1. Change in AI-factor loading (2023 to 2026 minus 2017 to 2022) against FY2023 data-centre mentions per 10,000 words, 33 US regulated electric utilities.

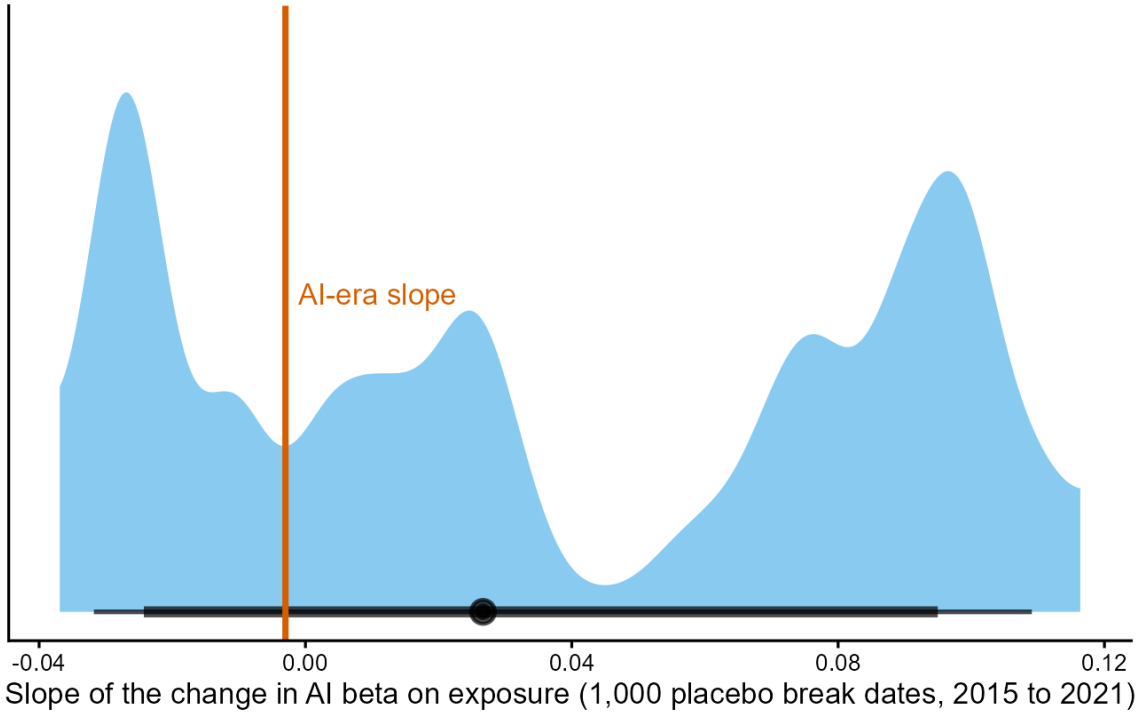


Figure 2. Exposure slope at 1,000 placebo break dates between 2015 and 2021, against the AI-era slope.

The levels say more than the changes. Regulated utilities load at about −0.28 on the AI factor before 2023 and −0.27 after: with the market held fixed, they rise when semiconductor stocks fall. Merchant generators moved from −0.32 to +0.46 on average: Vistra from −0.28 to +0.50, and Constellation, whose shares trade only from January 2022, from −0.36 (estimated on about ten months) to +0.41. Returns alone cannot show why, but the pattern fits investors pricing AI demand into generators that sell at market prices, and the regulated utilities that disclose the most data-centre activity show it as clearly as the rest.

5.2 Regulator-style betas fell

Table 2. Average beta of regulated electric utilities in each regulator's window

Country	Window	Utilities	Window to November 2022	Window to October 2026	Change
US	5 years, weekly	33	0.82	0.36	−0.46
UK	10 years, daily	2	0.76	0.76	−0.01
Germany	1 year, daily	1	0.71	0.40	−0.31
Spain	6 years, weekly	1	0.50	0.33	−0.17
Italy	5 years, daily	1	0.63	0.35	−0.27
Canada	5 years, weekly (descriptive)	4	0.60	0.35	−0.26

Table 2 shows that a regulator re-estimating beta in 2026 on its usual window would find a lower beta than in 2022 everywhere except the UK, whose 10-year window barely moved. Figure 3 splits each change into the three terms of the decomposition. The exposure and non-AI terms are negative in every country: utilities co-moved less with both parts of the index. The variance term is positive everywhere, because every earlier window contains the 2020 crash and market variance fell once it dropped out, so the

term records that fall. The exposure term is unrelated to data-centre exposure (US slope 0.04, standard error 0.10).

The design expected the variance term to be most negative where AI stocks dominate the index. That premise failed: the share of index variance explained by SMH fell in every country between the two windows, so the test of it (a correlation of −0.20 across six countries) carries little information.

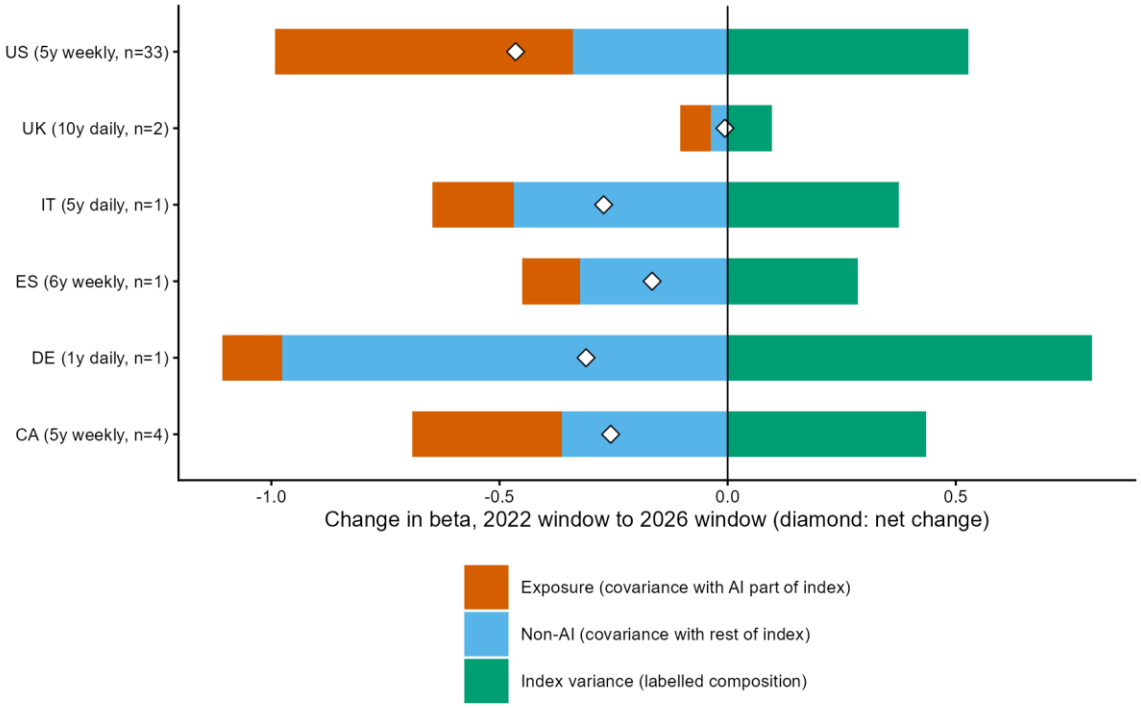


Figure 3. Decomposition of the change in regulator-style beta into exposure, non-AI and index-variance terms, by country.

5.3 The AI era added a few basis points to regulatory betas

Table 3. AI-era contribution to each determination

Determination	Replica check	AI-era contribution to beta	Effect on the comparator evidence
UK, RIIO-3 (10 years of daily data, assumed to end March 2025)	Gearing backed out from Ofgem's own comparator asset betas (mean 0.370; published 0.375), so not an independent reproduction; implied gearing plausible (0.29 to 0.55) for all eight comparators with a 10-year beta	+0.006 asset beta	+6 bp at 55% gearing and a 4.6% premium
Spain, 2026 to 2031 (6 years of weekly data to December 2023)	Levered betas correlate 0.99 with the CNMC's (mean gap 0.027); re-levered 0.78 against published 0.75	+0.013 equity beta	+6 bp at a 4.98% premium
Germany, prospective (windows ending October 2026)	6 of 12 comparators pass the gearing check on 1 and 3 years, all 12 on 5 years	+0.024 to +0.038 asset beta	+22 to +35 bp at 40% equity and a 3.7% premium
Italy, 2025 to 2027	Not replicated: comparator list	n/a	Betas unchanged by the 2024

	not published		update
Canada	No beta in the formula	Zero by construction	n/a

Table 3 and Figure 4 report the contribution for each determination. In every replicated case it is positive and small. The last column is the effect on the comparator evidence at the regulator's own parameters; it equals the effect on the allowed return only if the regulator's final choice moves one for one with the comparator average, which is not given where, as at Ofgem, the regulator picks a point within a range (0.30 to 0.45). Ofgem does not publish the cut-off of its window: ending it on 31 December 2024 or 30 September 2025 instead of 31 March 2025 gives 5.4 and 9.3 basis points (output/tables/h3_further_uk_end_dates.csv). The UK figure, 0.006, is equal to about a quarter of the 0.026 rise in Ofgem's asset beta between RIIO-2 (0.349) and RIIO-3 (0.375); the two numbers come from different calculations, and the rise also reflects new windows and comparators.

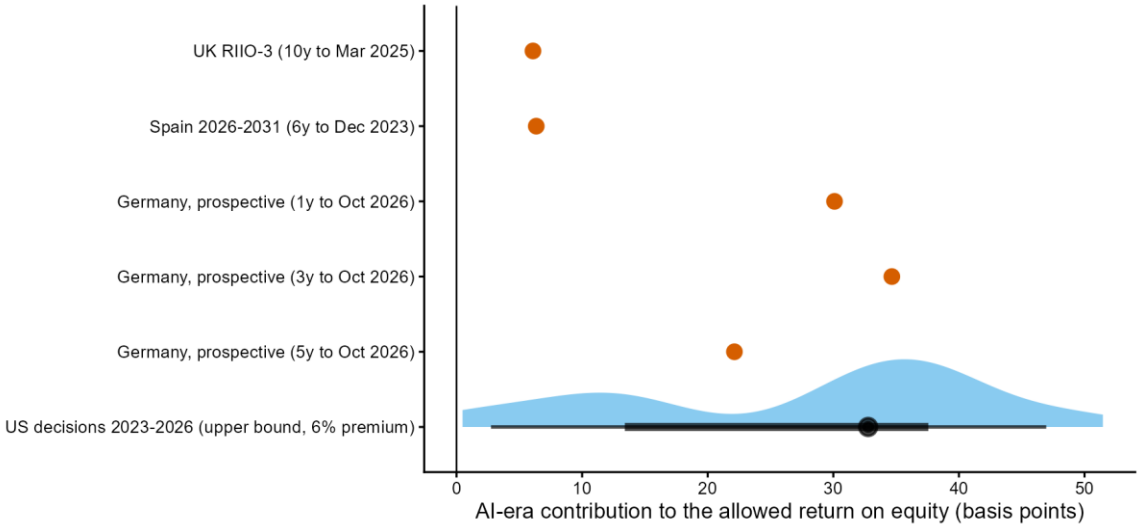


Figure 4. AI-era contribution to the allowed return on equity, in basis points, by determination and counterfactual.

The contribution comes almost entirely through the index. Putting each comparator's own AI loading back at its pre-AI value changes the beta by at most 0.002 in any determination. Resetting the index's slope on SMH does the rest. The slope fell in the AI era, so the counterfactual index moves more with semiconductor stocks than the actual one. For the S&P 500 the fall reflects SMH's own volatility (variance up 54%, covariance with the index up 11%, R^2 unchanged at 0.62); in the FTSE All-Share, the IBEX, the FTSE MIB and the STOXX index the covariance itself fell and the R^2 fell by 40% to 74% (output/tables/h3_further_moments.csv). Utilities move against the AI factor once the market is held fixed, so they co-move less with this more semiconductor-driven counterfactual index, and their counterfactual beta is lower. The actual beta exceeds it by the amounts in Table 3. The effect is mechanical: it reflects how the market proxy relates to semiconductor stocks, not a change in the risk of the regulated business.

The contributions are larger where more of the window falls in the AI era: 1 of 6 years for the CNMC, about 2.3 of 10 years for Ofgem, and all of a 1-year or 3-year German window ending in October 2026 (3.9 of 5 years for the 5-year window). With three determinations that also differ in frequency, premium, gearing and comparators, this is a pattern rather than a finding; the UK and Spain have different AI-era

shares and the same 6 basis points. Cross-border comparators carry part of the effect: of the UK's 6 basis points, 3.5 come from the four UK comparators and 2.5 from the four Spanish and Italian ones.

5.4 The contribution is not specific to the AI era

Table 4. AI-era contribution with bootstrap intervals and placebo ranks (basis points)

Determination	Slope held: contribution [95% bootstrap]	Placebo breaks at least as large	Variance share held: contribution [95% bootstrap]	Placebo breaks at least as large
UK RIIO-3	6.1 [1.5, 11.8]	87%	3.1 [0.0, 7.5]	53%
Spain 2026 to 2031	6.3 [2.8, 12.3]	5%	3.4 [1.0, 8.3]	52%
Germany, 1 year	30.1 [5.7, 59.7]	26%	12.3 [−5.8, 41.6]	20%
Germany, 3 years	34.7 [8.9, 62.6]	30%	16.3 [−6.7, 42.6]	1%
Germany, 5 years	22.1 [6.0, 38.0]	58%	11.9 [0.0, 26.3]	6%

1,000 month-block bootstrap draws; 1,000 placebo break dates between 2015 and 2021 (585 usable for Spain, whose comparator prices start in June 2017).

Table 4 and Figure 5 put each contribution against its sampling uncertainty and against placebo breaks. Given the break date, the main contributions are well measured: every bootstrap interval under the main counterfactual excludes zero. They are not unusual, however. Placebo break dates between 2015 and 2021, with the same window lengths and positions, give contributions at least as large in 87% of cases for the UK and 26% to 58% for the German windows; the placebo median for the UK is 10.8 basis points, above the AI-era value. Spain comes closest under the main counterfactual (5% of 585 usable breaks), but not under the variance-share counterfactual (52%). Holding the variance share fixed roughly halves every contribution, and the German 3-year window then stands out (1% of placebo breaks at least as large; 6% for 5 years, 20% for 1 year). The index's slope on SMH falls between the pre and post periods of 78% of placebo breaks for the S&P 500 and almost all for the FTSE All-Share, and SMH's variance rises after every one (output/tables/h3_further_slope_trend.csv), so a positive index-channel contribution of a few basis points is what a regulator would have found after almost any recent period.

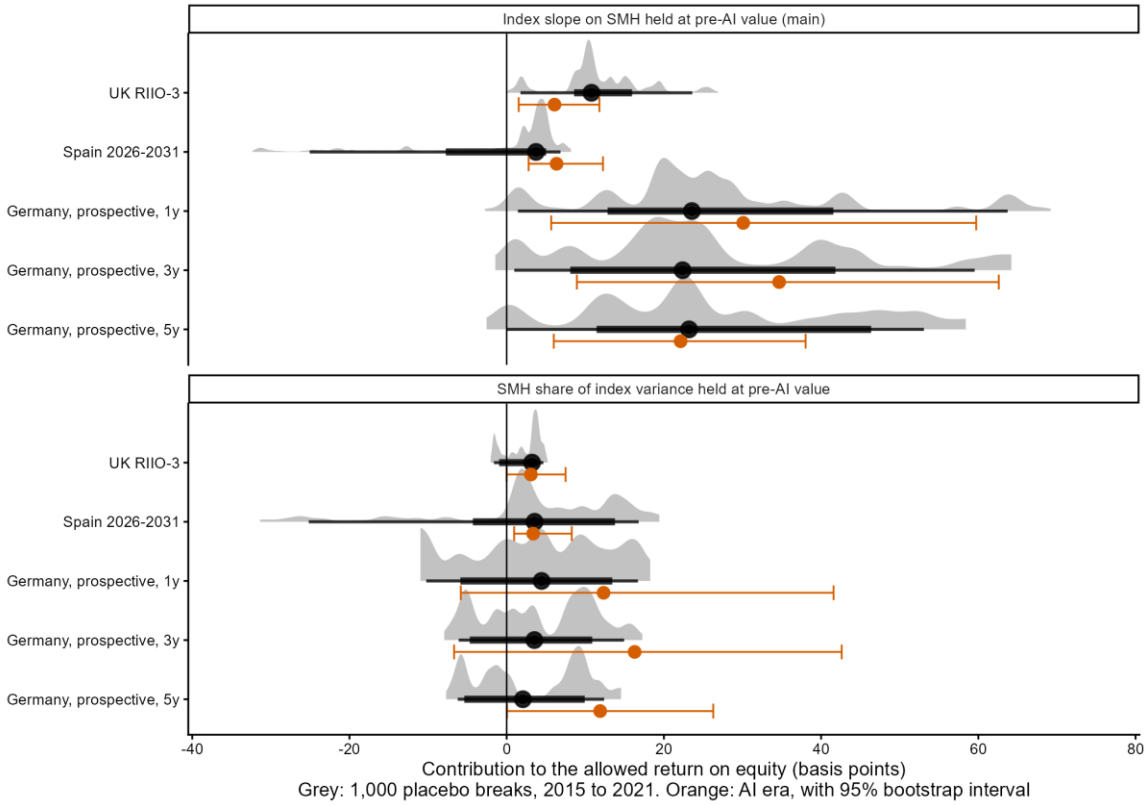


Figure 5. AI-era contribution (orange, with 95% month-block bootstrap interval) against contributions at 1,000 placebo break dates between 2015 and 2021 (grey), under the two composition counterfactuals.

5.5 US rate cases: no significant pass-through

At the 32 US decisions from 2023 on, the AI-era contribution to the Blume-adjusted beta averages +0.046 (range 0.001 to 0.086). Again it runs almost entirely through the index: −0.001 from the utilities' own AI loadings and +0.045 from the index.

Table 5. Change in authorised return on equity on the change in the AI-era contribution

Specification	Coefficient (return points per unit of beta)	Standard error	Wild-bootstrap p
State fixed effects (pre-specified)	2.65	1.74	0.29
No fixed effects	2.51	0.76	0.047
State and decision-year fixed effects	−3.40	6.91	0.62
Decision-year fixed effects	−0.12	2.32	0.96
AI-era share of window in place of c, state fixed effects	0.38	0.18	0.20

53 pairs of consecutive decisions, 18 states. Controls: changes in the 10-year Treasury yield and the Baa spread. Standard errors clustered by state. The pre-specification named clustering by utility; clustering by the 22 utilities with consecutive decisions gives standard errors of 1.51 (p = 0.08) for the pre-specified model and 0.83 (p = 0.01) without fixed effects (`output/tables/h3_further_us_clustering.csv`).

The pre-specified estimate in Table 5 is not distinguishable from zero with state clustering, and only marginally different from it with utility clustering (p = 0.08). Either way the design cannot test pass-through. The contribution is zero by construction at every decision before 2023, and after 2022 it is mostly a function of the decision date: the share of the 5-year window that falls after November 2022 explains 80% of its variation across the 32 AI-era decisions, and the change in that share explains 90% of the change in c across the 53 pairs. The coefficient therefore compares return changes in pairs that span 2023 with those that do not. Once the year of the later decision is held fixed, nothing is left (−0.12, standard error 2.32). Under the variance-share counterfactual, the US contribution is close to zero (median −0.0004), because the share of S&P 500 variance explained by SMH did not change. Mean authorised returns were 9.78% before 2023 and 9.70% from 2023 on.

5.6 The money involved is small

In the UK, 6 basis points on RIIO-3 equity is about £0.27 million a year per £1 billion of regulatory asset value at 45% equity. For the latest AI-era decision of each of 24 US rate units, the estimated coefficient implies a median of 15 basis points, *7.3 million a year and* 5.60 per customer a year (maximum *12). Assuming full CAPM weight at a 6% premium, which gives an upper bound, the median is 34 basis points,* 16 million a year and *12.60 per customer (maximum* 27). These US figures use the FERC net-plant proxy for most units and so overstate the rate base. Spanish and German effects are reported in basis points only, because regulated asset bases were not collected.

6. Checks

Five of the seven pre-specified checks apply to the replicas; which ones, and why, was recorded before they ran. Table 6 reports the contribution in basis points.

Table 6. AI-era contribution under the pre-specified checks (basis points)

Determination	Main	Nvidia instead of SMH	Dollar returns	AI era from 2024	National Grid half US
UK RIIO-3	6.1	1.9	5.4	4.5	8.4
Spain 2026 to 2031	6.3	2.9	8.3	0	n/a
Germany, 1 year	30.1	10.2	32.1	32.1	n/a
Germany, 3 years	34.7	14.4	35.3	37.5	n/a
Germany, 5 years	22.1	8.9	19.0	21.6	n/a

The contribution is positive in every case except Spain with the AI era starting in 2024, where it is zero by construction because the CNMC's window ends in December 2023. Its size depends on the choice of AI factor: with Nvidia in place of SMH it falls by half to two thirds, because the indices' co-movement with Nvidia changed less than their co-movement with SMH.

In the US panel, the pre-specified coefficient is not significant in any check: 9.14 with Nvidia (p = 0.31), 1.94 with the AI era from 2024 (p = 0.37) and 1.00 excluding Edison International and PG&E, whose returns carry wildfire liabilities (p = 0.74). The specification without fixed effects, borderline in the main run, is not significant in any check either (p from 0.07 to 0.16). The Nvidia coefficient is larger because the Nvidia-based contribution is smaller (median 0.015 against 0.055), so the same return changes are divided by a smaller number. For H1, the exposure slope is small and insignificant in all six applicable checks, ranging from −0.022 to +0.028 (results_h1_h2.md).

7. What the results do not show

They do not show that AI raised utilities' cost of capital through data-centre demand: the exposure channel is close to zero in every country and every determination. Nor do they show that the AI era, rather than a longer trend in semiconductor volatility, produced the index-channel contribution: placebo break dates give contributions as large in most cases, and the contribution halves under a counterfactual that holds a different moment of the index fixed. Whether a regulator should net out an index-driven change of this kind is a policy question that the estimates cannot settle.

The US panel cannot identify pass-through. The regressor is mostly a function of the decision date, so the coefficient is confounded with any other post-2022 change in allowed returns, and with decision-year fixed effects it is zero. A test with more power would use decisions that differ in how much AI-era data their witnesses' betas contained at the same date, for example through the timing of filings within a year.

The exposure measure counts mentions in one annual report and may miss load that utilities describe elsewhere, and 33 utilities give little power to detect a small slope. Outside the US, each country has one or two listed regulated utilities.

The replicas are partial. Italy is not replicated because its comparator list is not published. The German calculation rests on six comparators for the 1 and 3-year windows, omits the Vasicek adjustment and is prospective, since the next electricity period starts in 2029. The UK and German checks back out gearing from the regulators' own asset betas rather than from independent balance-sheet data. Canadian formula parameters come from secondary sources. The US betas are measured against the S&P 500, while Value Line, the source most rate-case witnesses use, measures against the NYSE Composite; since the index channel carries the effect, the US contributions could differ under that index. The pre-specified spillover figures (Diebold and Yilmaz 2012) are not reported here.

8. Conclusion

A regulator that set beta from market data in 2025 or 2026 would have found a small change in its comparator evidence: about 6 basis points on the UK's and Spain's allowed returns for 2026 to 2031, and 22 to 35 basis points on a German estimate made now, when the index's slope on semiconductor returns is held at its pre-AI value. The change does not come from utilities' exposure to data centres, which left their co-movement with AI stocks unchanged. It comes from the market index, whose slope on semiconductor returns fell, and it halves when the semiconductor share of index variance is held fixed instead. Placebo break dates in 2015 to 2021 produce contributions as large in most cases, so the evidence does not tie it to the AI era. At the same time, regulator-style betas fell in five of the six countries, mostly because the 2020 crash dropped out of the windows, so the net pressure from current market evidence on the CAPM part of allowed returns has been downward.

For regulators, the money is small: about £0.27 million a year per £1 billion of UK asset value, and at most a median of $12.60 per US customer a year at full CAPM weight. The German result is the largest because its windows are short and recent. If semiconductor stocks' weight in market indices keeps changing, the window and the index a regulator chooses will decide how much of that change reaches allowed returns. BNetzA's next electricity determination, applying from 2029, offers an out-of-sample test: the test is to recompute the contribution on BNetzA's actual comparators and windows once they are published, which will end later than the October 2026 windows used here. For US rate cases, the next step

is the full set of witness betas in each docket, which would show directly how much of a commission's CAPM estimate the AI-era index channel reached.

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Code, data documentation and pre-specification: https://github.com/arianshiraz/ai-utility-cost-of-capital